

Velocity-Tracking Adaptive Sliding-Mode Controller: Stability, Disturbance Headroom, and Initial-Gain Sizing in the Julia Mecanum Simulator

Mecanum_MultiComponentFriction_SlipSpin_Roller_ASMC_3 (Julia)
Mecanum-PINN v12.5 design note — companion to the position-tracking analysis

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Abstract

This note analyses the *velocity-tracking* controller `asmc_torques_vel` as the inner loop of a cascade, fed by an outer world-frame position/heading loop, and uses the analysis to size the warm-start initial gains K_{i0} that the velocity gain law leaks toward. A first draft of this note treated the inner loop in isolation and implicitly assumed body-frame velocity tracking alone certifies the world-frame trajectory — it does not. As written, `asmc_torques_vel` chases body-frame setpoints that are open-loop in time; if the yaw rate tracks low, the heading error $\Delta\psi$ accumulates, the body axes point the wrong way in the world, and the world velocity $R(\psi)(V_x, V_y)^\top$ drifts even with all three body surfaces pinned at zero. §2 makes this failure explicit and shows why the cascade resolves it: the outer loop measures world position/heading, sees the drift, and re-commands the inner setpoints. The inner-loop relative-degree-one UUB proof (§§3–7) is unchanged — the surface dynamics collapse to $\dot{s}_i = -\tilde{K}_i \tanh(s_i/\varepsilon_i) + \tilde{d}_i$ directly — but the cascade adds two inner-loop disturbance terms: a setpoint-rate term and a heading-rotation (“circle”) term $\propto \dot{\psi} k_p e_{\text{pos}}$, both quantified in §2.4. The heading correction acts on the *shaped* error $e_\psi(d_\psi)$ of the position controller `asmc_torques`, not the raw residual; §2.6 gives the exact edits to `asmc_torques_vel`. For deployment and run-time-matched PINN data, §10 develops the preferred *mixed relative-degree* variant — degree-1 velocity surfaces on (x, y) , a degree-2 position-type surface $s_\psi = \dot{e}_\psi + \lambda_\psi e_\psi$ on yaw — which bounds the heading error inside a single ASMC (no outer loop, no separation caveat) and flags a $\sim 38\dot{\psi}$ N m yaw viscous term the present velocity controller omits. Timescale separation is comfortable: the inner loop runs at 25–170 rad/s, leaving a 5×-slower outer loop at ≤ 5 rad/s (translation) and ≤ 34 rad/s (yaw). Three numerical facts carry over: the equivalent control inverts the *augmented* M_{aug} (cross-coupling $\sim 13.4\%$, down from $\sim 17.7\%$); the smaller velocity caps $K_{\text{max}} = (60, 80, 100)$ N m make actuator saturation the binding (B1) constraint at the simultaneous-cap worst case; and the sign-indefinite leak-to- K_0 term is benign, pinning the relaxed gain at $0.95K_{i0}$. The sizing rule $K_{i0} \gtrsim D_{i,\text{op}}^{\text{wr}}/\tanh 1$ now includes the cascade terms in $D_{i,\text{op}}^{\text{wr}}$. All numerical values are verified against the exact platform parameters in the notebook.

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1 What changes relative to the position controller

The companion note analysed `asmc_torques`, in which the sliding surface is the standard first-order surface $s_i = \dot{e}_i + \lambda_i(e_i) e_i$ built on the *position* error, with a dynamic Gaussian $\lambda_i(e_i)$ and an equivalent control that cancels the $\dot{\lambda}$ feedforward. The velocity controller `asmc_torques_vel` is structurally simpler, and the simplifications all act in the direction of a *cleaner* proof:

1. **Relative degree one; no λ .** The surface is the body-frame velocity error directly,

$$s_x = V_x - V_x^d, \quad s_y = V_y - V_y^d, \quad s_\psi = \dot{\psi} - \omega^d, \quad (1)$$

read off lines `s_x = Vx - Vx_d`, etc. There is no $\dot{e} + \lambda e$ construction, no dynamic- λ Gaussian, and no $\dot{\lambda}$ term inside the equivalent control. The whole of §12 of the position note (“why the dynamic λ does not break the argument”) is vacuous here, and the λ -cancellation that produced the clean $\dot{s}$ in the position note is unnecessary — the cancellation is automatic.

2. **Equivalent control is pure acceleration feedforward.** The set-points are the body-frame desired accelerations directly, $A_{x,\text{eq}} = A_x^d$, $A_{y,\text{eq}} = A_y^d$, $\alpha_{\text{eq}} = \alpha^d$, from `get_Axb_des`, `get_Ayb_des`, `get_alpha_des_v`. There are *no* error-transport terms ($-\lambda\dot{e}$, $-\dot{\lambda}e$) and no desired-velocity Coriolis correction applied to the surface — the Coriolis, viscous, and centripetal wrench terms in $M_{x,\text{eq}}, M_{y,\text{eq}}, M_{\psi,\text{eq}}$ are otherwise identical to the position controller.
3. **The equivalent control inverts the augmented mass matrix.** The inverse-dynamics wrench uses $\tilde{m} = m_s + 4J_1/R^2$ and $I_{\psi,\text{aug}} = I_s + 4(l+h)^2 J_1/R^2$ (wheel rotational inertia reflected to the body), i.e. it inverts

$$M_{\text{aug}} = \begin{pmatrix} \tilde{m} & 0 & -ma_Y \\ 0 & \tilde{m} & ma_X \\ -ma_Y & ma_X & I_{\psi,\text{aug}} \end{pmatrix}, \quad (2)$$

not the bare platform matrix M of the position note. This is the matrix that appears in `M_aug` and in the MIMO-DOB compensation $\Delta M_c = -\text{diag}(R, R, 1) M_{\text{aug}} \hat{\delta}$.

4. **A third gain-law term.** The velocity gain law is

$$\dot{K}_i = \underbrace{\gamma_i \sigma_i b(K_i)}_{\text{growth}} - \underbrace{c_i \left(\frac{K_i}{K_{\max,i}} \right)^3}_{\text{cubic pushback}} - \underbrace{\sigma_i^* (K_i - 0.95 K_{i0}) e^{\kappa(1-s_i^2/(9\varepsilon_\psi^2))}}_{\text{leak toward } K_{i0}}, \quad (3)$$

with $\sigma_i \equiv s_i \tanh(s_i/\varepsilon_i) \geq 0$, smooth gate $b(K) = \frac{1}{2} - \frac{1}{2} \tanh(K - (K_{\max} - 2))$, and the velocity configuration (σ^* written for the leak coefficient to avoid clash with σ_i)

$$\begin{aligned} \gamma &= (8, 15, 25), \quad c = (0.1, 0.3, 0.5), \quad K_{\max} = (60, 80, 100) \text{ N m}, \\ \sigma^* &= (0.5, 0.25, 0.25), \quad \kappa = 0.25, \quad K_0 = (5, 5, 20) \text{ N m}, \quad \varepsilon = 0.05, \quad \varepsilon_\psi = 0.08. \end{aligned}$$

The leak term is new relative to the position note (which had only growth + cubic). It is sign-indefinite in $K_i - 0.95 K_{i0}$; we carry it through $\dot{V}$ in full in §4. Its fixed point is exactly the warm-start anchor $0.95 K_{i0}$, so sizing K_{i0} is the practical content of this note.

2 What body-frame velocity tracking does *not* certify

The UUB result proven below bounds the three body-frame surfaces s_x, s_y, s_ψ . It is tempting — and a first draft of this note did exactly this — to read that as “the platform follows the intended trajectory.” It does not. Two distinct gaps must be separated.

Gap 1 (benign): static pose offset. If the body-frame velocity is tracked with a bounded but nonzero heading offset $\Delta\psi$, the world velocity is $\dot{r}^{\text{world}} = R(\psi)(V_x, V_y)^\top = R(\psi_{\text{int}} - \Delta\psi)(V_x, V_y)^\top$, a *rotation* of the intended world velocity. The realised path is rotated, and possibly translated, relative to the plan. For PINN training this is irrelevant: the network is fed body-frame states $(V_x, V_y, \dot{\psi}, \omega_i)$ and forces, never world position, so a rotated/translated trajectory is the same training datum. A bounded $\Delta\psi$ is acceptable by design.

Gap 2 (the real problem): *growing* heading error changes the trajectory class. The damaging case is a $\Delta\psi$ that *grows with time*. A circle authored as a constant body velocity $(V_x^d, V_y^d) = (v, 0)$ with $\omega^d = v/\rho$ becomes, under accumulating heading error, a spiral — a qualitatively different trajectory with different excitation content. This is the concern that matters, and it lives entirely in the yaw channel.

2.1 The heading error is the time-integral of the yaw surface

The heading is the open-loop integral of yaw rate, and the intended heading is the integral of the commanded yaw rate, so

$$\Delta\psi(t) \equiv \psi(t) - \psi_{\text{int}}(t) = \int_0^t (\dot{\psi} - \omega^d) d\tau = \int_0^t s_\psi(\tau) d\tau. \quad (4)$$

This is the crux: **heading error is the integral of the yaw sliding variable**. UUB bounds $|s_\psi| \leq \bar{s}_\psi$, but says nothing about $\int s_\psi d\tau$. The worst-case bound is

$$|\Delta\psi(t)| \leq \int_0^t |s_\psi| d\tau \leq \bar{s}_\psi t, \quad (5)$$

which grows linearly. Whether $\Delta\psi$ actually drifts depends on the *mean* of the steady-state yaw residual:

- **Zero-mean** s_ψ (symmetric chatter about zero): the integral stays bounded; $\Delta\psi$ wanders but does not secularly grow. Acceptable.
- **Biased** s_ψ ($\langle s_\psi \rangle = \delta \neq 0$): then $\Delta\psi(t) \approx \delta t$ — secular drift, circle $\rightarrow$ spiral. Not acceptable.

2.2 What biases $\tilde{d}_\psi$, and hence s_ψ

Inside the boundary layer the yaw surface settles at

$$s_\psi^{\text{ss}} \approx \frac{\varepsilon_\psi D_\psi^{\text{wr}}}{K_\psi}, \quad (6)$$

(the β_ψ cancels through $D_\psi^{\text{eff}} = \beta_\psi D_\psi^{\text{wr}}$, exactly as in (25)). A *constant* component of the yaw disturbance D_ψ^{wr} therefore produces a *constant* s_ψ^{ss} , which integrates into linear heading drift. The yaw disturbance is generically *not* zero-mean:

- the multicomponent spin torque $M_{zi} \propto \chi^2$ does not average to zero over a sustained turn (the roller spin has a fixed sign while turning);
- roller-handoff transients in $\tilde{\varphi}_i$ are phase-locked to wheel angle, not symmetric in time;
- most importantly, the DOB-onset *structural asymmetry* already documented (translation compensation gain $\sim 2.25\times$ versus yaw lever $L_\psi \approx 0.13$) is a standing, signed offset on precisely this channel — the canonical DC bias source.

2.3 Quantifying the open-loop drift

With the operating yaw gain $K_\psi^{\text{eq}} \approx 79.4 \text{ N m}$ (cubic balance, §8), $\varepsilon_\psi = 0.08$, a representative biased yaw disturbance gives the following linear drift rates:

Biased D_ψ^{wr} (N m)	0.5	1.0	2.0
s_ψ^{ss} (mrad/s)	0.50	1.01	2.02
Heading drift over 30 s (deg)	0.87	1.73	3.47
Heading error per circle lap (deg, $v=0.3$, $\rho=2$, $T=41.9$ s)	1.21	2.42	4.84

The per-lap figure is the telling one: the error *accumulates each lap*, so a nominally closed circle opens into a spiral at $2\text{--}5^\circ$ per revolution. This is exactly the failure mode of concern, and it is invisible to the inner-loop UUB certificate.

2.4 The cascade closes the integral, using the shaped heading error

The fix is to feed the inner yaw-rate setpoint from an outer heading loop that *measures* ψ (which the simulator state carries at `state[4]`). To stay consistent with the position controller `asmc_torques`, the correction acts on the *shaped* heading error used there, not on the raw residual. With $d_\psi \equiv \psi - \psi_d$, define

$$e_\psi(d_\psi) = 2 \tan \frac{d_\psi}{2} (1 + 2(1 - \cos d_\psi)), \quad e'_\psi(d_\psi) = \sec^2 \frac{d_\psi}{2} (3 - 2 \cos d_\psi) + 4 \tan \frac{d_\psi}{2} \sin d_\psi, \quad (7)$$

exactly the `e_psi` and `edash_psi` of the position controller. The outer law is

$$\omega^d = \dot{\psi}_d + k_{p,\psi} e_\psi(d_\psi), \quad (\text{so } \omega^d > \dot{\psi}_d \text{ when } \psi \text{ leads } \psi_d). \quad (8)$$

Three properties of (7), all verified numerically on $(-\pi, \pi)$, make this a valid and well-behaved error coordinate:

1. *Sign-faithful*: $\text{sign } e_\psi(d_\psi) = \text{sign } d_\psi$, so the correction always pushes ψ toward ψ_d .
2. *Strictly monotone*: $e'_\psi(d_\psi) \geq e'_\psi(0) = 1 > 0$ everywhere on the domain, so there is no spurious equilibrium and the closed-loop pole never vanishes.
3. *Identity at small angle*: $e_\psi(d_\psi) = d_\psi + O(d_\psi^3)$ with slope exactly 1 at the origin; the shaping only steepens at large excursions ($e_\psi/d_\psi \approx 1.01$ at 0.1 rad, ≈ 1.10 at 0.3 rad), adding authority where the heading is far off without altering the near-zero behaviour that sets the steady-state drift.

Differentiating (4) (i.e. $\dot{d}_\psi = \dot{\psi} - \dot{\psi}_d$) and substituting (8),

$$\dot{d}_\psi = \dot{\psi} - \omega^d = \underbrace{(\dot{\psi} - \dot{\psi}_d)}_{\text{inner residual } s_\psi^{\text{in}}} - k_{p,\psi} e_\psi(d_\psi). \quad (9)$$

The accumulating integrator (4) is replaced by a *stable nonlinear first-order* system. Linearising about the equilibrium d_ψ^{ss} gives the pole $-k_{p,\psi} e'_\psi(d_\psi^{\text{ss}})$; since $e'_\psi \geq 1$, the shaped error is *never slower* than a raw-error design and is up to $\sim 30\%$ faster at large offsets. A biased inner residual no longer integrates without bound; it settles where $k_{p,\psi} e_\psi(d_\psi^{\text{ss}}) = \langle s_\psi^{\text{in}} \rangle$, i.e.

$$e_\psi(d_\psi^{\text{ss}}) = \frac{\langle s_\psi^{\text{in}} \rangle}{k_{p,\psi}} = \frac{\varepsilon_\psi D_\psi^{\text{wr}}}{k_{p,\psi} K_\psi} \implies |d_\psi^{\text{ss}}| \approx \frac{\varepsilon_\psi D_\psi^{\text{wr}}}{k_{p,\psi} K_\psi} \quad (10)$$

where the final approximation uses $e_\psi \approx d_\psi$ at the small settled angle. The resulting bounded heading offsets:

	$D_\psi^{\text{wr}} = 1.0$	$D_\psi^{\text{wr}} = 2.0$
$ d_\psi^{\text{ss}} $ at $k_{p,\psi} = 2$ (deg)	0.029	0.058
$ d_\psi^{\text{ss}} $ at $k_{p,\psi} = 5$ (deg)	0.012	0.023
$ d_\psi^{\text{ss}} $ at $k_{p,\psi} = 10$ (deg)	0.006	0.012

The drift is now a fixed sub-tenth-of-a-degree offset (Gap 1, benign), not a secular growth (Gap 2); the exact- e_ψ form gives essentially the same steady bound as a raw correction because d_ψ^{ss} is tiny, but inherits the position controller's superior large-angle recovery. Optionally, the inner equivalent control can cancel the rate of the correction, $\dot{\omega}^d = \dot{\psi}_d + k_{p,\psi} e'_\psi(d_\psi) \dot{d}_\psi$, by setting $\alpha_{\text{eq}} = \alpha_{\text{ff}} - k_{p,\psi} e'_\psi(d_\psi) s_\psi$ (using $\dot{d}_\psi \approx s_\psi$); without it the un-cancelled rate is the small setpoint-rate disturbance quantified below. The same construction applies to the translational channels if world position must be bounded — $(V_x^d, V_y^d) \leftarrow R(\psi)^\top (\dot{r}_d^{\text{world}} + K_p e_{\text{pos}}^{\text{world}})$ — but per the stated training use only the heading channel needs closing; a bounded translational offset is acceptable.

2.5 Timescale separation and the cost to the inner loop

The cascade is valid only if the outer loop is slower than the inner. The inner boundary-layer linearisation $\dot{s}_i \approx -(\beta_i K_i / \varepsilon_i) s_i$ gives inner bandwidths

$$\omega_i^{\text{in}} = \frac{\beta_i K_i^{\text{eq}}}{\varepsilon_i} \approx (24.8, 28.2, 171.3) \text{ rad/s}, \quad (11)$$

so a $5\times$ separation permits outer gains up to $\omega_i^{\text{in}}/5 \approx (5.0, 5.6, 34.3) \text{ rad/s}$ — the yaw loop in particular can be made fast. The cascade does add two terms to the inner-loop disturbance, both small:

1. *Setpoint-rate term.* The outer correction makes ω^d time-varying; if the inner feedforward α_{eq} does not include $\dot{\omega}^d = k_{p,\psi} e'_{\psi}(d_{\psi}) \dot{d}_{\psi}$, the un-cancelled rate leaks into $\tilde{d}_{\psi}$. Bounded by $k_{p,\psi} e'_{\psi}(d_{\psi}) \bar{s}_{\psi}$, i.e. a few percent of the existing budget at the separations above (the $e'_{\psi} \rightarrow 1$ near steady state, so this is the same order as the raw-error estimate).
2. *Heading-rotation (“circle”) term.* For the translational channels, if closed, the rotation $R(\psi)^{\top}$ applied to a world correction injects a body-frame setpoint rate $\dot{\psi} k_p e_{\text{pos}}$; at $\dot{\psi} = 0.3 \text{ rad/s}$, $k_p = 1$, $e_{\text{pos}} = 0.1 \text{ m}$ this is 0.03 m/s^2 , i.e. $\sim 1.35 \text{ N m}$ of wrench — negligible against the $\sim 45\text{--}76 \text{ N m}$ headroom.

Both are folded into $D_{i,\text{op}}^{\text{wr}}$ for the K_{i0} sizing of §9; neither threatens the (B1) margin.

2.6 Implementation in `asmc_torques_vel`

The cascade requires three local edits to `asmc_torques_vel` and one new field on `ASMCPARAMS`; the inner relative-degree-one structure is untouched.

1. **Unpack the heading.** Add $\psi = \text{state}[4]$ to the state unpack (currently only `state[1:3, 17:19]` are read).
2. **Corrected yaw-rate setpoint.** Replace the open-loop ω_d with the shaped-error cascade (8): with $\omega_{\text{ff}} = \text{get_omega_des_v}(t)$, $\psi_d = \text{get_psi_des}(t)$, $d_{\psi} = \psi - \psi_d$, e_{ψ} from (7),

$$\omega_d = \omega_{\text{ff}} + \text{asmc.kp_psi} \cdot e_{\psi}(d_{\psi}), \quad s_{\psi} = \dot{\psi} - \omega_d.$$

3. **(Optional) feed-forward the correction rate.** Replace $\alpha_{\text{eq}} = \text{get_alpha_des_v}(t)$ with $\alpha_{\text{eq}} = \alpha_{\text{ff}} - \text{asmc.kp_psi} \cdot e'_{\psi}(d_{\psi}) s_{\psi}$, using e'_{ψ} from (7). This cancels the setpoint-rate disturbance; omitting it leaves the small leak of item 1 above.

The new parameter is a single field `kp_psi::Float64 = 5.0` (rad/s) on `ASMCPARAMS`. The choice $k_{p,\psi} = 5$ sits at $\sim \omega_{\psi}^{\text{in}}/34$, far inside the $5\times$ separation margin, and yields $|d_{\psi}^{\text{ss}}| \lesssim 0.02^\circ$. Two consistency requirements on the reference, both satisfied by the existing symbolic construction ($\psi_{\text{sym}} = \pi/2 + \psi_{\text{silu}}$, $\omega_{\text{sym}} = \dot{\psi}_{\text{sym}}$): (i) `get_psi_des` must be the exact integral of the yaw-rate feedforward that `get_omega_des_v` returns, or the correction fights the feedforward; (ii) at $t = 0$, $\psi(0) = \psi_d(0)$ makes $e_{\psi}(0) = 0$, so the correction is inert at start and adds no startup transient.

3 Plant on the sliding manifold and the canonical form

With $v = (V_x, V_y, \dot{\psi})^\top$, the body-frame dynamics actually used by the velocity controller read

$$M_{\text{aug}} \dot{v} = \tau_{\text{task}} + \mathcal{F}_{\text{fric}}(\chi, \mu, v, \omega, \theta) + \mathcal{C}(v), \quad (12)$$

where $\tau_{\text{task}} = (M_x^{\text{cmd}}, M_y^{\text{cmd}}, M_\psi^{\text{cmd}})^\top$ is the wrench delivered after the wheel-map and tanh saturation, and $\mathcal{F}_{\text{fric}}$ collects the multicomponent LuGre/Adamov forces and spin torques. Because the equivalent control is precisely $M_{\text{aug}} A_{\text{eq}} - \mathcal{C}(v) - \mathcal{F}_{\text{fric}}^{\text{model}}$, the closed loop on the nominal model is

$$\dot{v} = A_{\text{eq}} + M_{\text{aug}}^{-1} \tau_{\text{sw}} + M_{\text{aug}}^{-1} d(t), \quad (13)$$

where $d(t)$ lumps friction-model mismatch (the χ -dependent spin torque M_{zi} in particular), parametric error in $(m_s, I_s, J_1, p_1, p_2, a_X, a_Y)$, actuator-saturation residual, roller-handoff transients, and any state-estimation error if run open-loop on a PINN.

Differentiating the surface (1) and substituting (13), with $A_{\text{eq}} = \dot{v}^d$ (the desired body acceleration),

$$\dot{s}_i = [\dot{v}]_i - [\dot{v}^d]_i = [M_{\text{aug}}^{-1} \tau_{\text{sw}}]_i + [M_{\text{aug}}^{-1} d]_i. \quad (14)$$

There is no pairwise λ -cancellation to perform: relative degree one means the feedforward $A_{\text{eq}} = \dot{v}^d$ cancels the desired-acceleration term exactly, and (14) is immediate. This is the structural payoff of velocity tracking — the same canonical $\dot{s}$ the position note had to *engineer* via $\dot{\lambda}$ inside $A_{i,\text{eq}}$ falls out for free.

Remark 1 (Channel coupling through M_{aug}^{-1} ; the velocity numbers). With the exact notebook parameters ($h = 0.235$, $l = 0.15$, $R = 0.05$, $R_d = 0.0355$, $m = 30$, $m_w = 1.4$, $J_1 = 5.87 \times 10^{-3}$, $m_s = 35.6$, $I_s = 4.42$, so $\tilde{m} = 44.99$, $I_{\psi,\text{aug}} = 5.812$, $ma_X = 0.48$, $-ma_Y = 0.78$):

$$M_{\text{aug}} = \begin{pmatrix} 44.992 & 0 & 0.780 \\ 0 & 44.992 & 0.480 \\ 0.780 & 0.480 & 5.812 \end{pmatrix}, \quad M_{\text{aug}}^{-1} \approx \begin{pmatrix} 2.228 \times 10^{-2} & 3.19 \times 10^{-5} & -2.992 \times 10^{-3} \\ 3.19 \times 10^{-5} & 2.225 \times 10^{-2} & -1.841 \times 10^{-3} \\ -2.992 \times 10^{-3} & -1.841 \times 10^{-3} & 1.726 \times 10^{-1} \end{pmatrix}. \quad (15)$$

The smallest eigenvalue of M_{aug} is $\approx 5.79 > 0$, so M_{aug} is symmetric positive definite and the diagonal projection coefficients $\beta_i \leq [M_{\text{aug}}^{-1}]_{ii}$ are bounded away from zero. As fractions of the diagonal,

$$\frac{\sum_{j \neq x} |[M_{\text{aug}}^{-1}]_{xj}|}{[M_{\text{aug}}^{-1}]_{xx}} \approx 13.6\%, \quad \frac{\sum_{j \neq y} |[M_{\text{aug}}^{-1}]_{yj}|}{[M_{\text{aug}}^{-1}]_{yy}} \approx 8.4\%, \quad \frac{\sum_{j \neq \psi} |[M_{\text{aug}}^{-1}]_{\psi j}|}{[M_{\text{aug}}^{-1}]_{\psi \psi}} \approx 2.8\%.$$

The worst coupling is again $x \leftrightarrow \psi$, but at 13.4% it is *lower* than the position note's 17.7%: reflecting the wheel inertia into the body inflates the diagonal $\tilde{m}$ and $I_{\psi,\text{aug}}$ faster than the off-diagonals ma_X, ma_Y (which are unchanged), shrinking the relative coupling. The two devices of the position note carry over verbatim:

- *Device 1.* Absorb cross-axis switching into the disturbance. Since $|\tanh(s_j/\varepsilon_j)| \leq 1$ and the gate keeps $K_j \leq K_{\text{max}} + O(c/\gamma)$,

$$D_i^{\text{eff}} = D_i^{\text{wr}} \cdot \beta_i + \sum_{j \neq i} |[M_{\text{aug}}^{-1}]_{ij}| K_{\text{max},j}, \quad (16)$$

where D_i^{wr} is the genuine model-mismatch in wrench units.

- *Device 2.* Define $\beta_x \leq 0.0223$, $\beta_y \leq 0.0222$, $\beta_\psi \leq 0.1726$ (a safe practical choice $\beta_x = \beta_y = 0.020$, $\beta_\psi = 0.15$). With $\tilde{K}_i = \beta_i K_i$,

$$\dot{s}_i = -\tilde{K}_i \tanh(s_i/\varepsilon_i) + \tilde{d}_i, \quad |\tilde{d}_i| \leq D_i^{\text{eff}}, \quad (17)$$

the canonical scalar form.

4 Lyapunov function and its derivative

4.1 Candidate

Identical to the position note. For each axis fix a nominal ideal gain $K_i^* > 0$ and use the gain-error rescaling that places β_i in the numerator,

$$V_i(s_i, K_i) = \frac{1}{2}s_i^2 + \frac{\beta_i}{2\gamma_i}(K_i - K_i^*)^2, \quad V = \sum_i V_i. \quad (18)$$

4.2 Derivative, carrying the leak term in full

Differentiating (18) along (17) and (3), with $\sigma_i \equiv s_i \tanh(s_i/\varepsilon_i) \geq 0$ and writing $g_i \equiv e^{\kappa(1-s_i^2/(9\varepsilon_\psi^2))} > 0$ for the (bounded, positive) leak modulation:

$$\begin{aligned} \dot{V}_i &= s_i \dot{s}_i + \frac{\beta_i}{\gamma_i}(K_i - K_i^*)\dot{K}_i \\ &= -\beta_i K_i \sigma_i + s_i \tilde{d}_i + \frac{\beta_i}{\gamma_i}(K_i - K_i^*) \left[\gamma_i \sigma_i b(K_i) - c_i \left(\frac{K_i}{K_{\max,i}} \right)^3 - \sigma_i^* (K_i - 0.95K_{i0}) g_i \right]. \end{aligned} \quad (19)$$

The first two bracketed pieces reproduce the position note exactly. Grouping the $K_i \sigma_i$ terms via the β_i -match, $-\beta_i K_i \sigma_i + \beta_i K_i \sigma_i b(K_i) = -\beta_i K_i \sigma_i (1 - b(K_i))$,

$$\dot{V}_i = -\beta_i [K_i \sigma_i (1 - b(K_i)) + K_i^* \sigma_i b(K_i)] + s_i \tilde{d}_i - \frac{\beta_i c_i}{\gamma_i} (K_i - K_i^*) \left(\frac{K_i}{K_{\max,i}} \right)^3 - \underbrace{\frac{\beta_i \sigma_i^*}{\gamma_i} (K_i - K_i^*) (K_i - 0.95K_{i0}) g_i}_{\text{leak contribution } L_i}. \quad (20)$$

4.3 The leak term is benign

$L_i = -\frac{\beta_i \sigma_i^*}{\gamma_i} (K_i - K_i^*) (K_i - 0.95K_{i0}) g_i$ is a *product of two affine factors* in K_i , with the positive weight $\beta_i \sigma_i^* g_i / \gamma_i$. Two cases:

- If K_i^* and $0.95K_{i0}$ lie on the same side of K_i , the two factors share a sign and $L_i \leq 0$: the leak *helps* $\dot{V}_i$.
- Otherwise $L_i > 0$ but is uniformly bounded. The product of affine factors is maximised at the midpoint $K_i = \frac{1}{2}(K_i^* + 0.95K_{i0})$, giving

$$L_i \leq \frac{\beta_i \sigma_i^* \bar{g}_i}{4\gamma_i} (K_i^* - 0.95K_{i0})^2, \quad \bar{g}_i = e^\kappa = e^{0.25} \approx 1.284, \quad (21)$$

using $g_i \leq e^\kappa$ (the exponent is largest at $s_i = 0$). This is a fixed additive constant, exactly like the cubic-pushback residual: it enlarges the ultimate ball but does not threaten sign-definiteness of the dominant $-\beta_i K_i^* \sigma_i$ term.

Numerically, with $\sigma_x^* = 0.5$, $\gamma_x = 8$, $\beta_x = 0.020$, and a generous gap $|K_x^* - 0.95K_{x0}| \leq 50$,

$$L_x \leq \frac{0.020 \cdot 0.5 \cdot 1.284}{4 \cdot 8} (50)^2 \approx 1.0 \text{ (m/s}^2 \text{ units)},$$

comparable to the cubic-pushback residual and small against the dominant term over the operating range. The leak's *equilibrium* (where its own contribution to $\dot{K}_i$ vanishes) is $K_i = 0.95K_{i0}$, which is the warm-start anchor.

5 Region 1: adaptation active ($K_i \ll K_{\max} - 2$)

Here $b(K_i) \approx 1$, so $1 - b(K_i) \approx 0$ and (20) reduces to

$$\dot{V}_i \leq \underbrace{-\beta_i K_i^* \sigma_i + s_i \tilde{d}_i + \frac{\beta_i c_i}{\gamma_i} K_i^* \left(\frac{K_i^*}{K_{\max}} \right)^3}_{\leq \beta_i c_i K_{\max} / \gamma_i} + \underbrace{\frac{\beta_i \sigma_i^* \tilde{g}_i}{4\gamma_i} (K_i^* - 0.95K_{i0})^2}_{\text{leak residual (21)}}. \quad (22)$$

Outside the boundary layer ($|s_i| \geq \varepsilon$) the smoothed sign gives $\sigma_i \geq |s_i| \tanh 1 \approx 0.7616|s_i|$, hence

$$\dot{V}_i \leq -(\beta_i \tanh 1 K_i^* - D_i^{\text{eff}})|s_i| + (\text{const}). \quad (23)$$

The first term is strictly negative when $\beta_i \tanh 1 K_i^* > D_i^{\text{eff}}$. Using $D_i^{\text{eff}} = \beta_i D_i^{\text{wr}}$ (both $\dot{s}_i$ and $\tilde{d}_i$ are projected by the same row of M_{aug}^{-1}), the β_i cancels and the condition is on the *raw* gain:

$$K_i^* > \frac{D_i^{\text{wr}}}{\tanh 1} \approx 1.313 D_i^{\text{wr}}$$

(24)

— identical in form to the position note. The leak residual and cubic residual both enter only as fixed additive constants, so they set the size of the ultimate ball, not its existence. Inside the boundary layer $\tanh(s_i/\varepsilon) \approx s_i/\varepsilon$ and the surface linearises to $\dot{s}_i \approx -\frac{\beta_i K_i}{\varepsilon} s_i + \tilde{d}_i$, attracting s_i to

$$|s_i|_{\text{ss}} \lesssim \frac{\varepsilon D_i^{\text{wr}}}{K_i}, \quad (25)$$

with β_i again cancelling through $D_i^{\text{eff}} = \beta_i D_i^{\text{wr}}$.

6 Region 2: gain saturating ($K_i \rightarrow K_{\max}$)

As K_i passes $K_{\max} - 2$, $b(K_i) \rightarrow 0$ and the growth term shuts off. From (3) the gain rolls back via cubic pushback and leak, $\dot{K}_i \approx -c_i(K_i/K_{\max})^3 - \sigma_i^*(K_i - 0.95K_{i0})g_i < 0$, so the cap is asymptotic, not penetrated. With $b \approx 0$ in (20) the first bracket becomes $K_i \sigma_i \geq (K_{\max} - 2)\sigma_i$, and outside the boundary layer

$$\dot{V}_i \leq -(\beta_i \tanh 1 (K_{\max} - 2) - D_i^{\text{eff}})|s_i| + (\text{const}), \quad (26)$$

negative provided $D_i^{\text{wr}} < \tanh 1 (K_{\max} - 2)$. Region 2 is therefore the *safest* region for stability; its cost is sustained actuator saturation and chatter, which is exactly what the cubic pushback and leak exist to avoid.

7 Uniform ultimate boundedness

Theorem 1 (UUB of (s, K) for the velocity controller). *Under the bounded-disturbance assumption $||M_{\text{aug}}^{-1}d||_i \leq D_i^{\text{eff}}$, choose per axis*

$$K_{\max} > \frac{D_i^{\text{wr}}}{\tanh 1} \approx 1.313 D_i^{\text{wr}}, \quad K_i^* \in \left(\frac{D_i^{\text{wr}}}{\tanh 1}, K_{\max} - 2 \right), \quad (27)$$

and any $K_i(0) \geq 0$. Then there is a ball

$$\mathcal{B} = \left\{ (s, K) : |s_i| \leq \varepsilon + O(D_i^{\text{wr}}/K_{\max}), 0 \leq K_i \leq K_{\max} + O((c_i + \sigma_i^*)/\gamma_i) \right\}$$

that every trajectory of (17),(3) enters in finite time and never leaves.

Sketch. V is positive definite, radially unbounded, C^1 . Outside the boundary layer in Region 1, summing (22) over axes gives $\dot{V} \leq -\sum_i \beta_i (\tanh 1 K_i^* - D_i^{\text{wr}}) |s_i| + \sum_i (\beta_i c_i K_{\max}/\gamma_i + L_i^{\text{max}})$; the first sum is strictly negative under (24) and the second is a fixed positive constant, so V enters and stays in a sublevel set. Inside the boundary layer $|s_i| \leq \varepsilon$ bounds s directly; Region 2 bounds the K -component because both cubic pushback and the leak prevent K_i from exceeding $K_{\max}$ by more than $O((c_i + \sigma_i^*)/\gamma_i)$. UUB follows by the standard Lyapunov–Yoshizawa argument. $\square$

8 Disturbance headroom for the velocity configuration

The admissible task-space disturbance per axis is $D_i^{\text{wr}} < \tanh 1 \cdot K_{\max}$. With the velocity caps $K_{\max} = (60, 80, 100) \text{ N m}$,

Axis i	x	y	ψ
$K_{\max}$ (velocity, no-DOB)	60	80	100
Admissible $D_i^{\text{wr}} = \tanh 1 \cdot K_{\max}$ (N m)	45.7	60.9	76.2
Cross-coupling cost (Rem. 1, at $K_{\max}$)	8.1	6.7	2.8

8.1 The binding constraint moves to actuator saturation

The dominant high-gain disturbance is the actuator-saturation residual (B5 of the position note). The per-wheel switching torque after the wheel-map, in the worst-case aligned configuration with all three gains at the cap and all surfaces outside their boundary layers, is

$$|M^{\text{sw, wheel}}| \leq \frac{1}{4} (K_{\max}^x + K_{\max}^y + L_{\psi} K_{\max}^{\psi}) \tanh 1 \approx 0.25(60 + 80 + 0.13 \cdot 100)(0.762) \approx 29.1 \text{ N m},$$

with $L_{\psi} = R/(l+h) \approx 0.13$. Against $M_{\max} = 10 \text{ N m}$ the per-wheel residual is $29.1 - 10 \tanh(2.91) \approx 19.2 \text{ N m}$, projecting to $\Delta D^{\text{sat}} \leq 4 \times 19.2 \approx 76.7 \text{ N m}$ on a single task axis.

Remark 2 (Saturation overruns x -axis headroom at the simultaneous-cap worst case). This is the one place the velocity configuration is materially *tighter* than the position case. With the position note’s $K_{\max} = 100$ on all axes the headroom was 76.2 N m everywhere; the velocity x -axis cap of 60 drops its headroom to 45.7 N m , while the worst-case saturation residual is essentially unchanged at $\sim 77 \text{ N m}$. The naive worst-case budget on x therefore reads

$$D_x^{\text{wr}}|_{\text{worst}} \approx \underbrace{8.1}_{\text{cross}} + \underbrace{2}_{\text{fric}} + \underbrace{5}_{\text{Coriolis/param}} + \underbrace{76.7}_{\text{sat}} \approx 91.9 \text{ N m} > 45.7 \text{ N m},$$

a *negative* margin. The resolution is the same as in the position note: this combination — all three K_i pinned at the cap *and* all three surfaces outside their boundary layers *simultaneously* — does not persist in operation:

- The cubic-pushback/leak equilibrium pins K_i at $K_i^{\text{eq}} = K_{\text{max}}(\gamma_i \eta_i / c_i)^{1/3}$ for chatter η_i , below the cap.
- Inside the boundary layer $\tanh(s_i/\varepsilon) \approx s_i/\varepsilon \ll 1$, so the switching wrench is far below K_i and saturation never triggers.
- Saturation is a transient of the reaching phase only.

The practical conclusion is sharper than for the position controller: **with the velocity caps, the x -axis must not be driven outside its boundary layer with K_x near the cap.** This is precisely why K_{x0} should be sized to the *operating* disturbance (§9), keeping the reaching phase short and K_x well below the cap, rather than to the worst case.

9 Initial-gain (K_0) sizing

The leak term (3) drives K_i toward $0.95K_{i0}$ whenever the growth term is quiet (small σ_i). So K_{i0} is the *operating gain the controller relaxes to*, and it must by itself dominate the *operating* (not worst-case) disturbance, or the surface escapes its boundary layer before adaptation can grow K_i . The relevant budget excludes the saturation residual (which does not fire in the boundary layer) and uses the actual small- K cross-coupling (which scales with the operating K , not K_{max}). For velocity tracking the operating disturbance is friction-mismatch + Coriolis/parametric, on the order of

$$D_{x,\text{op}}^{\text{wr}} \approx 5, \quad D_{y,\text{op}}^{\text{wr}} \approx 5, \quad D_{\psi,\text{op}}^{\text{wr}} \approx 2 \text{ N m}$$

(the ψ -channel friction spin torque for $\chi = 8 \text{ mm}$, $\mu = 0.6$, $N_i \approx 74 \text{ N}$, $V_P \approx 0.1 \text{ m/s}$ is $M_{zi} \approx 5.7 \times 10^{-3} \text{ N m/wheel}$, i.e. $\sim 3 \times 10^{-3} \text{ N m}$ on the heading channel — negligible; the 2 N m is the parametric/Coriolis floor). Applying the Region-1 reaching condition (24) at startup,

$$K_{i0} \gtrsim \frac{D_{i,\text{op}}^{\text{wr}}}{\tanh 1} \approx 1.313 D_{i,\text{op}}^{\text{wr}} \quad (28)$$

Axis i	x	y	ψ
Operating $D_{i,\text{op}}^{\text{wr}}$ (N m)	5.0	5.0	2.0
Lower bound $K_{i0} \geq D_{i,\text{op}}^{\text{wr}} / \tanh 1$ (N m)	6.6	6.6	2.6
Current notebook setting K_{i0}	5	5	20
Equilibrium $K_i^{\text{eq}} = K_{\text{max}}(\gamma_i \eta / c_i)^{1/3}$, $\eta = 0.01$	55.7	63.5	79.4

Reading the table.

- x, y : the current $K_0 = 5$ is marginally *below* the 6.6 N m floor. At exactly 5 the startup surface can drift outside the boundary layer for the first $\sim 50 \text{ ms}$ until adaptation lifts K . Raising to $K_{x0} = K_{y0} = 8\text{--}10$ buys a clean $\sim 1.5\text{--}2\times$ margin over the operating disturbance with no downside (the cubic/leak still cap the steady gain at K^{eq}).

- ψ : the current $K_{\psi 0} = 20$ is far *above* the 2.6 N m floor — a deliberate, defensible warm start. Heading needs aggressive initial authority to suppress the DOB-onset transient (the structural translation/yaw asymmetry: translation compensation gain $\sim 2.25\times$ vs. yaw lever $L_{\psi} \approx 0.13$). Sitting $K_{\psi 0}$ near $0.2K_{\max}$ pre-loads the heading loop so the reaching phase is essentially instantaneous, at the cost of a little startup chatter — exactly the trade the leak term is tuned for. The same asymmetry is the DC bias source that drives the heading drift of §2: a high $K_{\psi 0}$ shrinks the steady-state $s_{\psi}^{\text{ss}} = \varepsilon_{\psi} D_{\psi}^{\text{wr}} / K_{\psi}$ and so reduces the open-loop drift rate, but it cannot eliminate it — only the outer heading loop (8) does that. No change recommended, but $K_{\psi 0}$ is not a substitute for closing the cascade.

Cascade terms in the operating budget. The two cascade disturbances of §2.4 — the setpoint-rate term ($\leq k_{p,\psi} \bar{s}_{\psi}$, a few percent) and the heading-rotation term (~ 1.35 N m on translation) — are already inside the $D_{i,\text{op}}^{\text{wr}}$ figures above; neither moves the K_{i0} floors materially.

Reaching time at startup. From the quadratic balance $t_{\text{reach}} \lesssim D_i^{\text{wr}} / (\gamma_i |s_i(0)|)$, with $|s_i(0)| \sim$ the initial velocity error: for $D^{\text{wr}} \sim 5$ N m, $\gamma_x = 8$, $|s_x(0)| \sim 0.5$ m/s, reaching the boundary layer takes $\sim 5/(8 \cdot 0.5) \approx 1.25$ s if starting from $K = 0$ — but only ~ 0.2 s when warm-started at $K_{x0} = 8$, which already nearly satisfies (24). This is the concrete payoff of sizing K_0 correctly.

10 Mixed relative-degree variant: degree-1 translation, degree-2 yaw

The cascade of §2.4 bounds heading with an *outer* loop. An alternative — preferred for deployment and for generating PINN data that matches the controller actually used at run time — folds the heading objective *into the ASMC itself* by giving the yaw axis its natural relative degree. The translational axes stay exactly as analysed (degree 1, velocity error); only the yaw surface changes:

$$s_x = V_x - V_x^d, \quad s_y = V_y - V_y^d, \quad \boxed{s_{\psi} = \dot{e}_{\psi} + \lambda_{\psi}(e_{\psi}) e_{\psi}}, \quad (29)$$

with e_{ψ} the shaped heading error (7) and $\dot{e}_{\psi} = e'_{\psi}(d_{\psi})(\dot{\psi} - \dot{\psi}_d)$. This is the *exact* yaw surface of `asmc_torques`; the mixed controller is “velocity ASMC on (x, y) , position ASMC on ψ .” The relative-degree structure now matches the control objective axis-by-axis: translation wants velocity (degree 1, position free), heading wants the angle (degree 2). The asymmetry is correct, not a hack.

10.1 Why this kills the drift without an integrator

The degree-1 yaw residual integrated into drift, $\Delta\psi = \int s_{\psi} d\tau$ (4). With (29), the manifold $s_{\psi} = 0$ is itself the stable first-order law $\dot{e}_{\psi} = -\lambda_{\psi} e_{\psi}$, so $e_{\psi} \rightarrow 0$ *on the surface*: a biased $\tilde{d}_{\psi}$ no longer integrates, it is absorbed into a bounded steady error exactly as in the position note,

$$|e_{\psi}|_{\text{ss}} \lesssim \frac{s_{\psi}^{\text{ss}}}{\lambda_{\psi, \min}} = \frac{\varepsilon_{\psi} D_{\psi}^{\text{wr}}}{\lambda_{\psi, \min} K_{\psi}}. \quad (30)$$

	$D_{\psi}^{\text{wr}} = 1.0$	$D_{\psi}^{\text{wr}} = 2.0$
$ e_{\psi} _{\text{ss}}$ at $\lambda_{\psi, \min} = 0.5$ (deg)	0.116	0.231
$ e_{\psi} _{\text{ss}}$ at $\lambda_{\psi, \max} = 5.0$ (deg)	0.012	0.023

The dynamic Gaussian $\lambda_\psi(e_\psi)$ rises toward $\lambda_{\psi,\max}$ as $e_\psi \rightarrow 0$, so the *operating* bound is the tight right column ($\leq 0.02^\circ$); the worst case uses $\lambda_{\psi,\min}$. Either way it is a fixed offset (Gap 1, benign), versus the degree-1 secular drift of $1.7\text{--}3.5^\circ$ over 30 s. No $k_{p,\psi}$, no timescale-separation assumption — λ_ψ does the job a single self-contained certificate.

10.2 What the proof inherits, and what changes

The aggregate Lyapunov function becomes *mixed*: the (x, y) blocks keep the relative-degree-1 canonical form of §§3–7 verbatim, while the ψ block reverts to the position note’s degree-2 analysis — the dynamic- λ cancellation $\dot{s}_\psi = [M_{\text{aug}}^{-1}\tau_{sw}]_\psi + [M_{\text{aug}}^{-1}d]_\psi$ holds because the yaw equivalent control again carries the $\dot{\lambda}_\psi$ feedforward (below). The per-axis β_i -cancellation and the (B1) bound $K_{\max} > D_\psi^{\text{wr}}/\tanh 1$ are unchanged; one simply can no longer claim “uniformly relative degree 1.” Two genuine shifts:

- *Reduced-order heading dynamics.* On the manifold the heading settles with time constant $1/\lambda_\psi \in [0.2, 2.0]$ s. The reaching (boundary-layer) bandwidth $\beta_\psi K_\psi/\varepsilon_\psi \approx 171$ rad/s is unchanged.
- *Cross-coupling carries a position-type term.* The yaw switching $K_\psi \tanh(s_\psi/\varepsilon_\psi)$ still leaks into x via $[M_{\text{aug}}^{-1}]_{x\psi}$, bounded by $|[M_{\text{aug}}^{-1}]_{x\psi}|K_\psi \approx 0.30$ N m (0.7% of the x headroom) because $|\tanh| \leq 1$ — the Device-1 bound is identical. But s_ψ now contains $\lambda_\psi e_\psi$ (a position quantity), so during large heading transients the leak sits near its bound more of the time; the *operating* D_x^{wr} during turns is marginally higher, still negligible against the ~ 46 N m x headroom.

10.3 The yaw equivalent control must be restored in full

Degree-2 yaw tracking requires the $\dot{\lambda}_\psi$ feedforward, identical to `asmc_torques`:

$$\alpha_{\text{eq}} = \alpha_{\text{ff}} - \dot{\lambda}_\psi e_\psi - \lambda_\psi \dot{e}_\psi e'_\psi, \quad (31)$$

where $(\lambda_\psi, \dot{\lambda}_\psi) = \text{get_dynamic_lambda}(e_\psi, e'_\psi \dot{e}_\psi, \lambda_{\psi,\min}, \lambda_{\psi,\max}, \mu_\psi)$ and $\dot{e}_\psi = e'_\psi(\dot{\psi} - \dot{\psi}_d)$. **One correction that matters independently of the surface choice:** the present `asmc_torques_vel` yaw equivalent wrench omits the yaw viscous-drag term that `asmc_torques` carries,

$$\left(\frac{4p_1(l+h)^2}{R^2} + \frac{8p_2h^2}{(R-R_d)^2} \right) \dot{\psi} \approx 38.2 \dot{\psi} \text{ N m}, \quad (32)$$

i.e. ~ 11.5 N m at $\dot{\psi} = 0.3$ rad/s. Left uncanceled this is a large, $\dot{\psi}$ -proportional component of $\tilde{d}_\psi$ — it would *worsen* the very drift the mixed surface fixes. Restoring it (it is a known, exactly-cancellable drag) shrinks D_ψ^{wr} materially and should be done whether or not the surface is changed.

10.4 Code edits for the mixed controller

Relative to the current `asmc_torques_vel`, the edits are confined to the yaw channel; (x, y) are untouched.

1. **Unpack pose and read the desired heading.** Add $\psi = \text{state}[4]$; compute $d_\psi = \psi - \text{get_psi_des}(t)$, then e_ψ and e'_ψ from (7) (the `e_psi/edash_psi` expressions).
2. **Degree-2 yaw surface.** Replace $s_\psi = \dot{\psi} - \omega_d$ with $\dot{e}_\psi = e'_\psi(\dot{\psi} - \text{get_omega_des_v}(t))$ and $s_\psi = \dot{e}_\psi + \lambda_\psi e_\psi$, where $(\lambda_\psi, \dot{\lambda}_\psi) = \text{get_dynamic_lambda}(e_\psi, e'_\psi \cdot (\dot{\psi} - \omega_{\text{ff}}), \dots)$. Leave s_x, s_y exactly as they are.

3. **Yaw equivalent control (31).** Replace $\alpha_{\text{eq}} = \text{get_alpha_des_v}(\mathbf{t})$ with $\alpha_{\text{eq}} = \alpha_{\text{ff}} - \dot{\lambda}_{\psi} e_{\psi} - \lambda_{\psi} \dot{e}_{\psi} e'_{\psi}$.
4. **Restore the yaw viscous term (32)** by adding $+(4p_1(l+h)^2/R^2 + 8p_2h^2/(R-R_d)^2)\dot{\psi}$ inside $M_{\text{psi_eq}}$, matching `asmc_torques`.
5. **Parameters.** The yaw λ fields already exist on `ASMCParams` ($\lambda_{\psi,\text{min}}=0.5$, $\lambda_{\psi,\text{max}}=5.0$, $\mu_{\psi}=100$); no `kp_psi` is needed (the cascade gain is retired). The adaptive-gain block and the (x, y) surfaces are unchanged.

At $t = 0$, $\psi(0) = \psi_d(0) \Rightarrow e_{\psi}(0) = 0$ and $\lambda_{\psi}e_{\psi}(0) = 0$, so the yaw surface starts on its manifold with no startup transient, provided $\dot{\psi}(0) = \omega_{\text{ff}}(0)$ (already true by construction).

10.5 PINN-data implication

The mixed controller changes the yaw excitation the network sees: s_{ψ} carries $\lambda_{\psi}e_{\psi}$, so the heading-channel torque spectrum on turns gains low-frequency content (the $\lambda_{\psi}e_{\psi}$ term) absent from the degree-1 data. This is the controller that will run under MPC, so it is the correct distribution to train on — but datasets generated with the degree-1 velocity controller and the mixed controller must *not* be pooled; regenerate consistently.

11 Conclusion

The central result of this revision is a correction: body-frame velocity-error UUB does *not* certify the intended trajectory. A bounded pose offset is harmless for PINN training (Gap 1), but a *growing* heading error (Gap 2) changes the trajectory class — a circle opens into a spiral. Because heading error is the time-integral of the yaw surface, $\Delta\psi = \int s_{\psi} d\tau$, any DC bias in the yaw disturbance (the χ^2 spin torque, phase-locked roller handoff, and above all the 2.25:0.13 DOB translation/yaw asymmetry) integrates into linear drift — quantified here at 2–5° per circle lap. The inner-loop UUB proof is silent on this. The fix is the cascade: an outer heading loop $\omega^d = \dot{\psi}_d + k_{p,\psi}e_{\psi}(d_{\psi})$, acting on the position controller’s *shaped* heading error e_{ψ} , converts the unbounded integrator into a stable nonlinear first-order system, bounding d_{ψ} at a fixed sub-0.1° offset. Because $e'_{\psi} \geq 1$ everywhere, the shaped form is never slower than a raw-error correction and recovers faster from large excursions. Timescale separation is comfortable (inner 25–171 rad/s; outer ≤ 5 –34 rad/s), and the cascade’s added disturbance is negligible against the (B1) headroom.

For deployment and for PINN data that matches the run-time controller, the *mixed relative-degree* variant (§10) is preferred: keep the degree-1 velocity surfaces on (x, y) but give yaw its natural degree-2 surface $s_{\psi} = \dot{e}_{\psi} + \lambda_{\psi}e_{\psi}$ on the shaped error. The manifold itself then enforces $\dot{e}_{\psi} = -\lambda_{\psi}e_{\psi}$, so a biased $\tilde{d}_{\psi}$ produces a *bounded* $|e_{\psi}|_{\text{ss}} \leq \varepsilon_{\psi} D_{\psi}^{\text{wr}} / (\lambda_{\psi,\text{min}} K_{\psi}) \lesssim 0.02$ – 0.23° rather than secular drift — no outer loop, no timescale-separation caveat. The cost is a mixed certificate (degree-1 on (x, y) , degree-2 on ψ) and the restored yaw equivalent control, including the $\sim 38.2 \dot{\psi}$ N m viscous term (32) the present velocity controller omits — a correction worth making on its own merits, since uncanceled it is itself an ~ 11 N m yaw disturbance.

The relative-degree-one inner-loop analysis itself transfers cleanly from the position note: no dynamic λ , no $\dot{\lambda}$ feedforward, the canonical $\dot{s} = -\tilde{K} \tanh(s/\varepsilon) + \tilde{d}$ falls out directly, and the β_i -cancellation and (B1) $K_{\text{max}} > D^{\text{wr}}/\tanh 1$ hold unchanged. Three numerical facts:

1. Inverting the augmented M_{aug} lowers worst-axis cross-coupling to $\sim 13.4\%$ (from $\sim 17.7\%$).

2. The smaller velocity caps make the simultaneous-cap actuator-saturation residual overrun the x -axis headroom; the design response is to keep K_x off the cap by sizing K_{x0} to the operating disturbance, not the worst case.
3. The leak-to- K_0 term is sign-indefinite but benign — a product of two affine factors bounded by (21) — pinning the relaxed gain at $0.95K_{i0}$.

The initial-gain rule $K_{i0} \gtrsim D_{i,\text{op}}^{\text{wr}} / \tanh 1$ recommends nudging K_{x0}, K_{y0} from 5 to $\sim 8\text{--}10\text{ N m}$ and confirms $K_{\psi 0} = 20$ as an intentional heading warm start — though a high $K_{\psi 0}$ only *reduces* the drift rate; closing the heading loop is what *bounds* it.

A Numerical scratchpad

$\tanh 1 = (e^2 - 1)/(e^2 + 1) = 0.76159\dots$ $L_\psi = R/(l + h) = 0.05/0.385 = 0.12987$. $\tilde{m} = m_s + 4J_1/R^2 = 35.6 + 4(5.87 \times 10^{-3})/0.0025 = 44.992$. $I_{\psi,\text{aug}} = I_s + 4(l + h)^2 J_1/R^2 = 4.42 + 4(0.385)^2(5.87 \times 10^{-3})/0.0025 = 5.812$. Smallest eigenvalue of $M_{\text{aug}} \approx 5.79$. $\beta_x \leq [M_{\text{aug}}^{-1}]_{xx} = 0.02228$, $\beta_y \leq 0.02225$, $\beta_\psi \leq 0.17261$. Leak modulation cap $\bar{g} = e^{0.25} = 1.2840$; leak vanishes at $K = 0.95K_0 = (4.75, 4.75, 19.0)$.